

LESSON 8.6

SOLVING MULTI-STEP EQUATIONS



LESSON INTRODUCTION

MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.

LEARNING TARGETS

- I can solve multi-step linear equations with variables on both sides.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

WORKING TOWARD

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

ESSENTIAL QUESTIONS

- How do you solve multi-step linear equations with variables on both sides?

LESSON NARRATIVE

This lesson focuses primarily on providing opportunities for students to practice solving multi-step equations in one variable. Students will engage in an error analysis before working with a partner to solve equations and sort them by the number of solutions each has. Consider ways to provide differentiated support to students who are still struggling to solve multi-step equations or to determine the number of solutions.

COMPONENT SUMMARY

8.6.1 WARM-UP (~5 MIN.)

Interpret data and statistics

8.6.2 COLLABORATION (10 MIN.)

Engage in Error Analysis solving a multi-step equation

8.6.3 ACTIVITY (25-30 MIN.)

Practice solving multi-step equations

8.6.4 WRAP-UP (~5 MIN.)

Reflect on understanding of learning targets in the unit so far

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- Equation Cards for Activity 8.6.3 (1 set per group)
- (Optional) Chart paper or drawing paper
- (Optional) Highlighters
- (Optional) Scientific calculator

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may apply the distributive property incorrectly when solving equations, including not distributing to all terms or failing to distribute a negative.
- Students may try to combine like terms across the equal sign.

THINGS TO CONSIDER

- Consider using information from the previous lessons to form a small group of students who are struggling solving multi-step equations and determining the number of solutions. Facilitate discussion with these students as they work through 8.6.3 Activity.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Consider implementing a Think-Pair-Share routine for the 8.6.3 Activity. This routine provides the opportunity for students to “take turns” performing error analysis and/or solving multi-step equations and then explaining and articulating their process with a partner.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

The 8.6.2 Collaboration provides the opportunity for students to engage in error analysis of solving a multi-step equation. Use this activity to provide students with the opportunity to examine their own thinking but to also learn from the thought processes of other students. Encourage students to explain how they engaged in error analysis and what they learned.

DIFFERENTIATE INSTRUCTION

Allow students to work the equations on scratch paper or chart paper if they struggle to write the equations in a small space. Consider making allowances for students who need more time by allowing students to solve fewer problems in the 8.6.3 Activity. Provide numerous ways for students to explain their responses to the Wrap-Up such as verbal description or providing examples.

8.6.1 WARM-UP: MATH ARGUMENT

TEACHER MOVES

Students may hesitate to respond to a written response question. Push students towards engagement by providing them sentence stems, such as:

- “I think Dave is a better player because his _____ is _____ than Michelle’s.”
- “I think Michelle is a better player because her _____ is _____ than Dave’s.”



8.6.1 Warm-Up: Math Argument

Two friends, Dave and Michelle, play Candy Crave often. Their records are shown below.

Dave’s Scores and Statistics		Michelle’s Scores and Statistics	
Games Played	529	Games Played	457
High Score	8,931	High Score	6,751
Average Score	5,126	Average Score	5,817
Latest Score	7,915	Latest Score	5,284
Longest Game	8 min 3 sec	Longest Game	5 min 9 sec
Total Play Time	12 hr 9 min	Total Play Time	9 hr 13 min

1. Use information from the table to argue why Dave is the better player.
Answers will vary. Dave has a higher high score. Dave’s latest score is higher than Michelle’s.
2. Use the same information to argue that Michelle is the better player instead.
Answers will vary. Michelle has a higher average score.



8.6.2 Collaboration: Finding the Error

1. Clare was solving an equation, but when she checked her solution, she noticed it was incorrect. She knows she made a mistake but cannot find it.

Clare's work is shown.

Clare's Work	Corrected Solution
$12(5 + 2y) = 4y - (5 - 9y)$	$12(5 + 2y) = 4y - (5 - 9y)$
$60 + 24y = 4y - 5 - 9y$	$60 + 24y = 4y - 5 + 9y$
$60 + 24y = -5y - 5$	$60 + 24y = 13y - 5$
$24y = -5y - 55$	$65 + 24y = 13y$
$29y = -55$	$65 = -11y$
$y = \frac{-55}{29}$	$-\frac{65}{11} = y$

- a. Discuss Clare's work with your partner to find her mistake. Then, circle where she made the mistake.
Claire did not distribute the negative sign to $-9y$. She also subtracted $-5 - 60$ incorrectly to arrive at -55 instead of -65 .
- b. Find the correct solution to the equation using the space provided above.
Shown above.
- c. Verify your solution is correct using substitution.

$$12\left(5 + 2\left(-\frac{65}{11}\right)\right) = -\frac{900}{11}$$

$$4\left(-\frac{65}{11}\right) - \left(5 - 9\left(-\frac{65}{11}\right)\right) = -\frac{900}{11}$$

8.6.2 COLLABORATION: FINDING THE ERROR

TEACHER MOVES

MA.K12.MTR.4.1: Discussion and Reflection

This activity allows students to engage in error analysis of solving a multi-step equation including the distributive property and combining like terms. Encourage students to explain the errors, why they occurred, and how they can be corrected.

TEACHER MOVES

Consider allowing students to use a scientific calculator to verify their solution after solving the equation. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator.

8.6.3 ACTIVITY: SOLUTION CARD SORT

TEACHER MOVES

Before beginning the activity, place students in pairs and give each pair a stack of 12 cards with equations for them to solve.

TEACHER MOVES

Each student will only solve six of the equations on their own. They will be independently solving different equations but can ask each other questions if they need support.

It should not be the expectation that students copy all of the work of their partner on their paper as well. Instead, students can share their solutions and verify each other's work before moving on to question 4 together.

TEACHER MOVES

Think-Pair-Share

Consider implementing a Think-Pair-Share routine for this activity. Encourage students to "take turns" solving each equation and then sharing and explaining their process with a partner. This routine gives students an opportunity to verbalize the problem-solving experience as well as to troubleshoot any issues they experienced.



8.6.3 Activity: Solution Card Sort

You and your partner will receive 12 cards for this activity.

1. Shuffle the cards and place them face-down in a pile.
2. Record your partner's name. _____
3. Each player draws a card from the pile and solves the equation in the corresponding box on the recording sheet below. Be sure to record the work in the correct box by matching the letter on the card to the letter in the box.

A $15g + 1 = -15(1 - g)$ $15g + 1 = -15 + 15g$ $1 \neq -15$ <i>No solution</i>	B $5 + 8x + x = 4x + 30$ $5 + 9x = 4x + 30$ $5 + 5x = 30$ $5x = 25$ $x = 5$
C $y - 3(2 - 3y) = 2(5y + 1)$ $y - 6 + 9y = 10y + 2$ $10y - 6 = 10y + 2$ $-6 \neq 2$ <i>No solution</i>	D $\frac{1}{4}(12 - 4d) = 3 - d$ $3 - d = 3 - d$ <i>Infinitely many solutions</i>
E $\frac{15t - 10}{5} = 3t - 2$ $15t - 10 = 15t - 10$ <i>Infinitely many solutions</i>	F $4(2c - 2) - 4 = 4(c - 2)$ $8c - 8 - 4 = 4c - 8$ $8c - 12 = 4c - 8$ $4c - 12 = -8$ $4c = 4$ $c = 1$

G $8 + 2n = -3.5n + 19$ $8 + 5.5n = 19$ $5.5n = 11$ $n = 2$	H $3(3j + 2) - 2j = 7j + 6$ $9j + 6 - 2j = 7j + 6$ $7j + 6 = 7j + 6$ <i>Infinitely many solutions</i>
I $-12 + 0.60(2e - 5) = 1.20e - 15$ $-12 + 1.2e - 3 = 1.2e - 15$ $-15 + 1.2e = 1.2e - 15$ <i>Infinitely many solutions</i>	J $\frac{1}{2}b - 3 = 2 - \frac{3}{4}b$ $\frac{5}{4}b - 3 = 2$ $\frac{5}{4}b = 5$ $b = 4$
K $0.80s - 3 = 7 + s - 0.20s$ $-0.8s - 3 = 7 - 0.8s$ $-3 \neq 7$ <i>No solution</i>	L $4(m - 1) = \frac{1}{2}(8m - 2)$ $4m - 4 = 4m - 1$ $-4 \neq -1$ <i>No solution</i>

4. When all equations have been solved, sort the cards into groups based on their number of solutions. Write the letters that correspond to the cards in each group in the table below.

One Solution	No Solution	Infinitely Many Solutions
<i>B, F, G, J</i>	<i>A, C, K, L</i>	<i>D, E, H, I</i>



8.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.
Answers will vary.



I need help.



I got it.



I could help someone.

I can solve multi-step linear equations.	  
I can solve equations with variables on both sides.	  
I can relate solutions to equations to real-world contexts they model.	  
I can recognize equations with one, none, or infinitely many solutions.	  

8.6.4 WRAP-UP: REFLECTION

TEACHER MOVES

In addition to having students circle an icon, encourage students to provide an example of specific issues that they still need help with.

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

TEACHER MOVES

Consider pairing students to discuss their reflections to see if peers can answer any of their questions.